

TASK 11

Topic : Library Management System

You are going to build a project based on Library Management System. It keeps track of all information about books in the library, their cost, status and total number of books available in the library.

Create a database named library and following TABLES in the database:

1. Branch
2. Employee
3. Books
4. Customer
5. IssueStatus
6. ReturnStatus

Attributes for the tables:

1. Branch

- Branch_no
- Set as PRIMARY KEY
- Manager_Id
- Branch_address
- Contact_no

```
1 • CREATE DATABASE library;
2 • USE library;
3
4 • ○ CREATE TABLE Branch (
5     Branch_no INT PRIMARY KEY,
6     Manager_Id INT,
7     Branch_address VARCHAR(100),
8     Contact_no VARCHAR(15)
9 );
```

```

56 • INSERT INTO Branch VALUES
57     (1, 101, 'Kozhikode', '9876543210'),
58     (2, 102, 'Malappuram', '9876543211'),
59     (3, 103, 'Ernakulam', '9876543212');
60
61 • SELECT * FROM Branch;

```

	Branch_no	Manager_Id	Branch_address	Contact_no
▶	1	101	Kozhikode	9876543210
	2	102	Malappuram	9876543211
	3	103	Ernakulam	9876543212
✱	NULL	NULL	NULL	NULL

2. Employee

- Emp_Id – Set as PRIMARY KEY
- Emp_name
- Position
- Salary
- Branch_no
- Set as FOREIGN KEY and it refer Branch_no in Branch table

```

11 • CREATE TABLE Employee (
12     Emp_Id INT PRIMARY KEY,
13     Emp_name VARCHAR(100),
14     Position VARCHAR(50),
15     Salary DECIMAL(10,2),
16     Branch_no INT,
17     FOREIGN KEY (Branch_no) REFERENCES Branch(Branch_no)
18 );

```

```

63 • INSERT INTO Employee VALUES
64     (101, 'Rahul', 'Manager', 60000, 1),
65     (102, 'Anjali', 'Manager', 65000, 2),
66     (103, 'Suresh', 'Manager', 70000, 3),
67     (104, 'Akhil', 'Clerk', 30000, 1),
68     (105, 'Meena', 'Assistant', 40000, 1),
69     (106, 'John', 'Clerk', 28000, 2),
70     (107, 'Priya', 'Assistant', 35000, 2),
71     (108, 'Arun', 'Clerk', 32000, 3);
72
73 • SELECT * FROM Employee;

```

	Emp_Id	Emp_name	Position	Salary	Branch_no
►	101	Rahul	Manager	60000.00	1
	102	Anjali	Manager	65000.00	2
	103	Suresh	Manager	70000.00	3
	104	Akhil	Clerk	30000.00	1
	105	Meena	Assistant	40000.00	1
	106	John	Clerk	28000.00	2
	107	Priya	Assistant	35000.00	2
	108	Arun	Clerk	32000.00	3
*	NULL	NULL	NULL	NULL	NULL

3. Books

- ISBN
- Set as PRIMARY KEY
- Book_title
- Category
- Rental_Price
- Status [Give yes if book available and no if book not available]
- Author
- Publisher

```

20 • CREATE TABLE Books (
21     ISBN VARCHAR(20) PRIMARY KEY,
22     Book_title VARCHAR(100),
23     Category VARCHAR(50),
24     Rental_Price DECIMAL(10,2),
25     Status VARCHAR(5),
26     Author VARCHAR(100),
27     Publisher VARCHAR(100)
28 );

75 • INSERT INTO Books VALUES
76 ('B101', 'World History', 'History', 30, 'yes', 'Arun Kumar', 'National Books'),
77 ('B102', 'Math Basics', 'Education', 20, 'yes', 'Ravi Sharma', 'Student Press'),
78 ('B103', 'Science Today', 'Science', 25, 'no', 'Anil Verma', 'Knowledge India'),
79 ('B104', 'Indian History', 'History', 35, 'yes', 'Meera Nair', 'Heritage Books'),
80 ('B105', 'English Grammar', 'Education', 15, 'yes', 'Sita Menon', 'Academic Press'),
81 ('B106', 'Physics Guide', 'Science', 40, 'no', 'Rajesh Gupta', 'Tech Books');
82
83 • SELECT * FROM Books;

```

	ISBN	Book_title	Category	Rental_Price	Status	Author	Publisher
►	B101	World History	History	30.00	yes	Arun Kumar	National Books
	B102	Math Basics	Education	20.00	yes	Ravi Sharma	Student Press
	B103	Science Today	Science	25.00	no	Anil Verma	Knowledge India
	B104	Indian History	History	35.00	yes	Meera Nair	Heritage Books
	B105	English Grammar	Education	15.00	yes	Sita Menon	Academic Press
	B106	Physics Guide	Science	40.00	no	Rajesh Gupta	Tech Books
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Customer

- Customer_Id
- Set as PRIMARY KEY
- Customer_name
- Customer_address
- Reg_date

```

30 • CREATE TABLE Customer (
31     Customer_Id INT PRIMARY KEY,
32     Customer_name VARCHAR(100),
33     Customer_address VARCHAR(100),
34     Reg_date DATE
35 );

85 • INSERT INTO Customer VALUES
86     (1, 'Aman', 'Kozhikode', '2021-05-10'),
87     (2, 'Nisha', 'Malappuram', '2023-03-15'),
88     (3, 'Ravi', 'Ernakulam', '2020-07-20'),
89     (4, 'Sneha', 'Kozhikode', '2022-01-10'),
90     (5, 'Fathima', 'Malappuram', '2021-12-01');
91
92 • SELECT * FROM Customer;

```

	Customer_Id	Customer_name	Customer_address	Reg_date
▶	1	Aman	Kozhikode	2021-05-10
	2	Nisha	Malappuram	2023-03-15
	3	Ravi	Ernakulam	2020-07-20
	4	Sneha	Kozhikode	2022-01-10
	5	Fathima	Malappuram	2021-12-01
*	NULL	NULL	NULL	NULL

5. IssueStatus

- Issue_Id
- Set as PRIMARY KEY
- Issued_cust – Set as FOREIGN KEY and it refer customer_id in CUSTOMER table
- Issued_book_name
- Issue_date
- Isbn_book – Set as FOREIGN KEY and it should refer isbn in BOOKS table

```

37 • CREATE TABLE IssueStatus (
38     Issue_Id INT PRIMARY KEY,
39     Issued_cust INT,
40     Issued_book_name VARCHAR(100),
41     Issue_date DATE,
42     Isbn_book VARCHAR(20),
43     FOREIGN KEY (Issued_cust) REFERENCES Customer(Customer_Id),
44     FOREIGN KEY (Isbn_book) REFERENCES Books(ISBN)
45 );

```

94 • INSERT INTO IssueStatus VALUES

```

95     (1, 1, 'World History', '2023-06-10', 'B101'),
96     (2, 2, 'Math Basics', '2023-06-15', 'B102'),
97     (3, 3, 'Indian History', '2023-05-20', 'B104'),
98     (4, 5, 'Physics Guide', '2023-06-18', 'B106');
99

```

100 • SELECT * FROM IssueStatus;

	Issue_Id	Issued_cust	Issued_book_name	Issue_date	Isbn_book
▶	1	1	World History	2023-06-10	B101
	2	2	Math Basics	2023-06-15	B102
	3	3	Indian History	2023-05-20	B104
	4	5	Physics Guide	2023-06-18	B106
✱	NULL	NULL	NULL	NULL	NULL

6. ReturnStatus

- Return_Id
- Set as PRIMARY KEY
- Return_cust
- Return_book_name
- Return_date
- Isbn_book2
- Set as FOREIGN KEY and it should refer isbn in BOOKS table

```

47 • CREATE TABLE ReturnStatus (
48     Return_Id INT PRIMARY KEY,
49     Return_cust INT,
50     Return_book_name VARCHAR(100),
51     Return_date DATE,
52     Isbn_book2 VARCHAR(20),
53     FOREIGN KEY (Isbn_book2) REFERENCES Books(ISBN)
54 );

102 • INSERT INTO ReturnStatus VALUES
103     (1, 1, 'World History', '2023-06-20', 'B101'),
104     (2, 2, 'Math Basics', '2023-06-25', 'B102');
105
106 • SELECT * FROM ReturnStatus;

```

	Return_Id	Return_cust	Return_book_name	Return_date	Isbn_book2
▶	1	1	World History	2023-06-20	B101
	2	2	Math Basics	2023-06-25	B102
•	NULL	NULL	NULL	NULL	NULL

Display all the tables and Write the queries for the following :

1. Retrieve the book title, category, and rental price of all available books.

```

108 • SELECT Book_title, Category, Rental_Price
109     FROM Books
110     WHERE Status = 'yes';

```

	Book_title	Category	Rental_Price
▶	World History	History	30.00
	Math Basics	Education	20.00
	Indian History	History	35.00
	English Grammar	Education	15.00

2. List the employee names and their respective salaries in descending order of salary.

```
112 • SELECT Emp_name, Salary
113 FROM Employee
114 ORDER BY Salary DESC;
```

	Emp_name	Salary
▶	Suresh	70000.00
	Anjali	65000.00
	Rahul	60000.00
	Meena	40000.00
	Priya	35000.00
	Arun	32000.00
	Akhil	30000.00
	John	28000.00

3. Retrieve the book titles and the corresponding customers who have issued those books.

```
116 • SELECT b.Book_title, c.Customer_name
117 FROM IssueStatus i
118 JOIN Books b ON i.Isbn_book = b.ISBN
119 JOIN Customer c ON i.Issued_cust = c.Customer_Id;
```

	Book_title	Customer_name
▶	World History	Aman
	Math Basics	Nisha
	Indian History	Ravi
	Physics Guide	Fathima

4. Display the total count of books in each category.

```
121 • SELECT Category, COUNT(*) AS total_books
122 FROM Books
123 GROUP BY Category;
```


	Category	total_books
▶	History	2
	Education	2
	Science	2

5. Retrieve the employee names and their positions for the employees whose salaries are above Rs.50,000.

```
125 • SELECT Emp_name, Position
126 FROM Employee
127 WHERE Salary > 50000;
```

	Emp_name	Position
▶	Rahul	Manager
	Anjali	Manager
	Suresh	Manager

6. List the customer names who registered before 2022-01-01 and have not issued any books yet.

```
129 • SELECT Customer_name
130 FROM Customer
131 WHERE Reg_date < '2022-01-01'
132 AND Customer_Id NOT IN (SELECT Issued_cust FROM IssueStatus);
```

	Customer_name
--	---------------

7. Display the branch numbers and the total count of employees in each branch.

```
134 • SELECT Branch_no, COUNT(*) AS total_employees
135 FROM Employee
136 GROUP BY Branch_no;
```

	Branch_no	total_employees
▶	1	3
	2	3
	3	2

8. Display the names of customers who have issued books in the month of June 2023.

```

138 • SELECT DISTINCT c.Customer_name
139     FROM IssueStatus i
140     JOIN Customer c ON i.Issued_cust = c.Customer_Id
141     WHERE MONTH(i.Issue_date) = 6 AND YEAR(i.Issue_date) = 2023;

```

	Customer_name
▶	Aman
	Nisha
	Fathima

9. Retrieve book_title from book table containing history.

```

143 • SELECT Book_title
144     FROM Books
145     WHERE Book_title LIKE '%history%';

```

	Book_title
▶	World History
	Indian History

10. Retrieve the branch numbers along with the count of employees for branches having more than 5 employees

```

147 • SELECT Branch_no, COUNT(*) AS total_employees
148     FROM Employee
149     GROUP BY Branch_no
150     HAVING COUNT(*) > 5;

```

	Branch_no	total_employees
--	-----------	-----------------

11. Retrieve the names of employees who manage branches and their respective branch addresses.

```

152 • SELECT e.Emp_name, b.Branch_address
153 FROM Employee e
154 JOIN Branch b ON e.Branch_no = b.Branch_no
155 WHERE e.Position = 'Manager';

```

	Emp_name	Branch_address
▶	Rahul	Kozhikode
	Anjali	Malappuram
	Suresh	Ernakulam

12. Display the names of customers who have issued books with a rental price higher than Rs. 25.

```

157 • SELECT DISTINCT c.Customer_name
158 FROM IssueStatus i
159 JOIN Customer c ON i.Issued_cust = c.Customer_Id
160 JOIN Books b ON i.Isbn_book = b.ISBN
161 WHERE b.Rental_Price > 25;

```

	Customer_name
▶	Aman
	Ravi
	Fathima